

Solution Requirements

Date	24 June 2026
Team ID	
Project Name	Human development index (HDI) Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system allows users (researchers, policymakers, and administrators) to securely access the application using username and password authentication	High
2	Authorization levels	Admins can manage datasets, retrain the ML model, and view reports. General users can input development indicators and obtain HDI	High

		predictions without modifying system data.	
3	External interfaces	The system integrates with CSV datasets, Python Machine Learning libraries (Scikit-learn, Pandas, NumPy), Flask web application, and visualization libraries (Matplotlib and Seaborn).	High
4	Transactions processing	The system accepts user inputs (Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI per capita), processes them through the trained ML model, and instantly predicts the HDI category and score.	High
5	Reporting	The system generates prediction results, visualizations, and summary reports showing HDI score, category, and comparison charts. Users can export prediction reports if required.	Medium
6	Business rules, etc.	All input values must be validated before prediction. Missing or invalid values are rejected. Predictions are generated only using the trained machine learning	High

		model and predefined HDI classification rules.	
7	Compliance to laws or regulations	The system complies with data privacy principles by using publicly available datasets, protecting user information, and ensuring secure handling of uploaded data.	Medium
8	<i>Other</i>	The system provides graphical visualizations of HDI indicators and supports future model retraining using updated datasets.	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	The application should generate HDI predictions quickly after user input	Prediction completed within 2 seconds .
2	Scalability	The system should support increasing datasets and multiple simultaneous users without performance degradation	Supports 500+ concurrent users and larger datasets.
3	Security & Data Privacy	User credentials and sensitive data should be securely stored and transmitted using encryption. Input validation should prevent malicious attacks.	Passwords encrypted, HTTPS enabled, protection against SQL Injection and XSS.
4	Reliability & Availability	The application should remain operational with minimal downtime and recover gracefully from failures.	99.9% monthly uptime with automatic error handling.
5	Usability & Accessibility	The web interface should be simple, responsive, and easy to use on desktops and mobile devices.	User can complete prediction in less than 3 steps . Responsive design with accessibility support.
6	<i>Other</i>	The code should be modular, documented, and easily deployable across Windows, Linux, and macOS platforms.	Modular Python code with documentation; deployable on multiple operating systems without modification.

